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Ridge Regression

Ridge regression is a linear regression technique that is used to handle multicollinearity, which occurs when independent variables in a regression model are highly correlated with each other. It is an extension of ordinary least squares (OLS) regression that introduces a regularization term to the regression equation.

In ridge regression, the sum of squared residuals, which is minimized in OLS regression, is modified by adding a penalty term known as the ridge penalty or L2 regularization term. The ridge penalty is calculated by multiplying the square of the coefficient values by a tuning parameter, typically denoted as λ . The larger the value of λ , the greater the impact of the ridge penalty.

By adding the ridge penalty to the regression equation, ridge regression reduces the coefficients of the correlated variables towards zero while still maintaining them in the model. This helps to reduce the impact of multicollinearity and stabilize the regression model. However, unlike variable selection techniques such as backward elimination or forward selection, ridge regression does not eliminate variables entirely but rather shrinks their coefficients towards zero.

The ridge penalty has the effect of reducing the variance of the regression coefficients, but it does not set them to exactly zero. This means that even less important variables can still have some influence on the response variable. The amount of shrinkage applied to the coefficients depends on the value of λ , with larger λ values resulting in more shrinkage.

Ridge regression is particularly useful when dealing with datasets that have a large number of correlated predictors. It helps to improve the predictive accuracy of the model by reducing the impact of multicollinearity and preventing overfitting. The choice of the optimal λ value is often determined through cross-validation techniques to strike a balance between model simplicity and performance. The following reading will give you a strong insight as to what Ridge Regression is and how it can be a powerful tool in machine learning.

[What is Ridge Regression in Machine Learning](#)

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